

8.3.1.1.5 Die Temperature Monitor

备注

When switching between external and internal temperature monitoring, a 2-s latency may be incurred due to the natural scheduler update interval.

A die temperature block generates a voltage that is proportional to the die temperature, and provides a way of reducing component count if pack thermistors are not used or ensuring that the die power dissipation requirements are observed. The die is measured using the same on-board 14-bit ADC as the cell voltages.

To select internal die temperature measurement mode, set $[TEMP_SEL] = 0$.

For BQ76930 and BQ76940, multiple die temperature measurements are available. These are stored in TS2 and TS3.

To convert a DIETEMP reading into temperature, refer to the following equation box. If more accurate temperature readings are needed from DIETEMP, the DIETEMP at room temperature value should be stored during production calibration.

The following equation shows how to use the 14-bit ADC readings in TS1, TS2, and TS3 when $[TEMPSEL] = 0$ to determine the internal die temperature:

$$V_{25} = 1.200 \text{ V (nominal)} \quad (6)$$

$$V_{TSX} = (\text{ADC in Decimal}) \times 382 \mu\text{V/LSB} \quad (7)$$

$$TEMP_{DIE} = 25^\circ - ((V_{TSX} - V_{25}) \div 0.0042) \quad (8)$$

8.3.1.1.6 16-Bit Pack Voltage

Once converted to digital form, each cell voltage is added up and the summation result stored in the BAT registers. The sum is divided by 4 so that the result of summing 15 cells fits in the 16-bit value. This 16-bit value has a nominal LSB of 1.532 mV.

The following shows how to convert the 16-bit pack voltage ADC reading into an analog voltage. This value also uses the GAIN and OFFSET stored into EEPROM.

The ADC transfer function is a linear equation defined as follows:

$$V_{(BAT)} = 4 \times \text{GAIN} \times \text{ADC}(\text{cell}) + (\# \text{Cells} \times \text{OFFSET}) \quad (9)$$

GAIN is stored in units of $\mu\text{V/LSB}$, while OFFSET is stored in mV units.

8.3.1.1.7 System Scheduler

A master scheduler oversees the monitoring intervals, creating a full update every 250 ms. Temperature measurements are taken every 2 seconds. Pack voltage is calculated every 250 ms. More information on the System Scheduler can be found in the [Embedded Scheduler in Cell Battery Monitor of the BQ769x0](#) application report.

8.3.1.2 Protection Subsystem

8.3.1.2.1 Integrated Hardware Protections

Integrated hardware protections are provided as an extra degree of safety and are meant to supplement the standard protection feature set that would be incorporated into the host controller firmware. They should not be used as the sole means of protecting a battery pack, but are useful for FMEA purposes; for example, in the event that a host microcontroller is unable to react to any of the below protection situations. All hardware protection thresholds and delays should be loaded into the AFE by the host microcontroller during system startup. The AFE will also default to predefined threshold and delay settings, in case the host microcontroller is unable to or does not wish to program the protection settings.